LAB 01 INTRODUCTION TO PROXMOX VE

Network Support Services | Guided technician study guide | Proxmox VE 9.x-style interface

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THE BIG IDEA

A single physical server can host multiple independent workloads. Proxmox VE is a Type 1 virtualization platform based on Debian Linux, KVM virtual machines, and Linux containers (LXC). Administrators manage nodes and guests through a web interface.

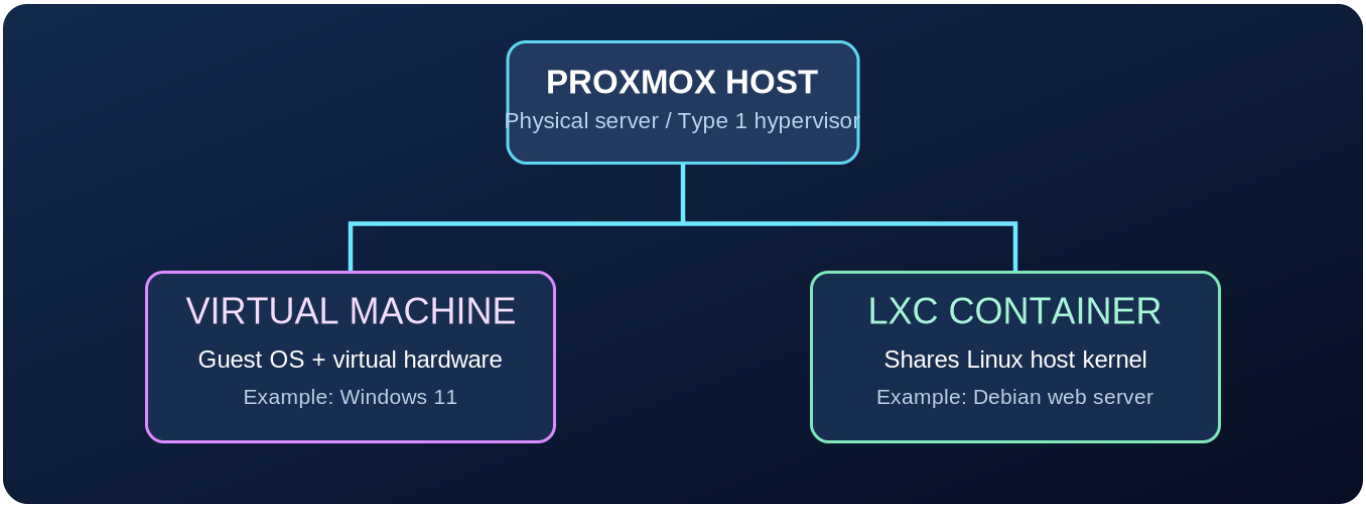


Figure 1. One physical host can run KVM virtual machines and Linux LXC containers. Illustrated training diagram, not an application screenshot.

Your mission

- Recognize hypervisors, nodes, VMs, containers, storage, and the Datacenter object.
- Use the mock console to inspect a sample host and identify CPU, memory and guest status.
- Complete the eight guided tasks, four GUI challenges, and six-question knowledge check.
- If authorized, compare the simulator with an actual Proxmox host using read-only views.

Passing: full student name, eight guided tasks, four verified GUI challenges, and at least 80/100 assessment points. Hardware verification is a separate student-attested section.

1 | Core virtualization concepts

Term	What a technician should know
Hypervisor (Type 1)	A platform that runs on physical server hardware and provides virtualized workloads.
Proxmox node	An individual Proxmox host participating in a datacenter or cluster.
Datacenter	Top-level Proxmox management context for nodes and global configuration.
KVM virtual machine	An isolated guest with virtual hardware and its own guest operating system kernel.
LXC container	An isolated Linux workload that shares the host Linux kernel.
vCPU / RAM	CPU execution and memory capacity assigned to a VM or container.
local storage	Often a directory storage target holding ISO images, templates and/or backups.
local-lvm	Often an LVM-thin pool holding VM and container disks; confirm actual configuration.

2 | How the interface is organized

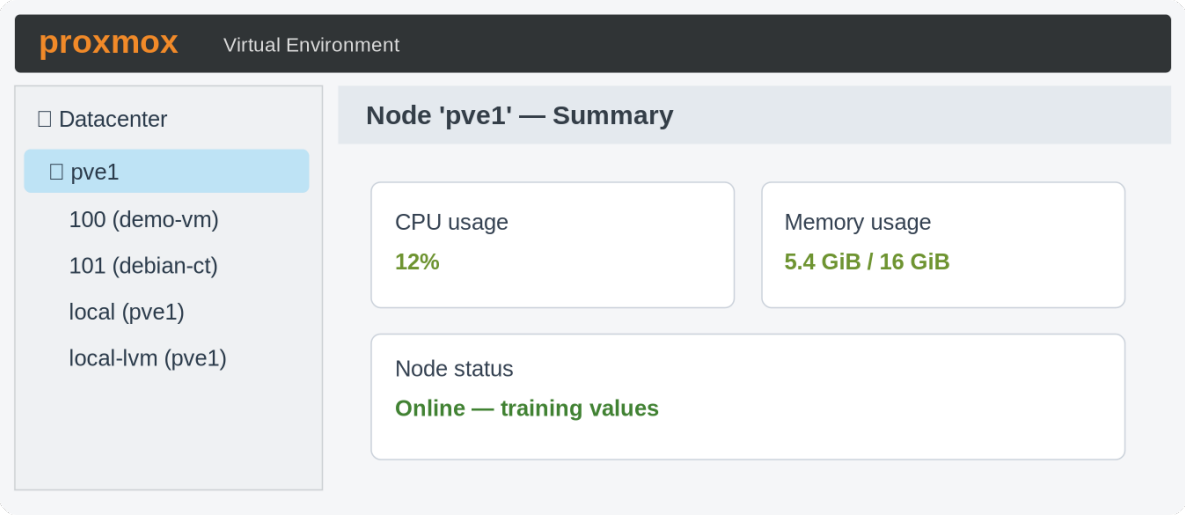


Figure 2. Stylized Proxmox console: the resource tree lists Datacenter, node, guests, and storage; the content pane displays selected settings.

- **Resource tree:** the hierarchy of datacenter, nodes, guest workloads, and storage.
- **Summary:** status and resource information for the selected node or workload.
- **Network:** interfaces and Linux bridges on a selected node.
- **Storage:** available disks/pools/content types; contents vary with configuration.
- **Shell:** node command line; requires care and authorized changes.

3 | Guided GUI inspection workflow

01. Identify the platform

Explain Type 1 hypervisors and why Proxmox can consolidate workloads.

02. Select Datacenter

Observe top-level resources, navigation and global settings.

03. Select pve1

Identify the simulated physical host and its Summary tab.

04. Review node health

Read CPU, RAM, and online status; recognize sample values as fictional.

05. Select VM 100

Observe a virtual machine with its own guest operating system.

06. Select CT 101

Compare LXC behavior with a KVM guest.

07. Compare storages

Compare local directory and local-lvm; inspect configured content types.

08. Explain observations

State why technicians check resources before adding new workloads.

4 | Four simulator scoring checkpoints

Challenge	Completion evidence	Points
Datacenter selected	Select the Datacenter object	10
Node Summary opened	Select pve1 and open Summary	10
VM and container inspected	Visit both guest resource views	10
Storage types inspected	Visit both local and local-lvm	10

Simulator checkpoints are automatically assessed inside the HTML lab. Guided checklist completion is tracked independently.

5 | Live node verification worksheet

Complete this optional section only on an instructor-authorized classroom host. Lab 01 is read-only. Do not reboot, erase, format, or change networking.

Step 1. Open an authorized management URL of the form **https://HOST-IP:8006**. A certificate warning can appear with an untrusted local certificate; ask your instructor before proceeding.

Step 2. Identify the node hostname and management IP.

Step 3. Open node Summary, record actual CPU/RAM use and online status.

Step 4. Identify existing storage labels and their purposes.

Node / IP: _____

CPU model or cores: _____

RAM use / total: _____

Storage targets: _____

Status / observations: _____

6 | Troubleshooting quick reference

Symptom	Safe first check
Browser cannot reach the web GUI	Verify authorized hostname/IP, port 8006, cabling, network connectivity and host status.
Unexpected login / certificate prompt	Verify you reached the correct host; ask the instructor about authentication and internal CA trust.
Node looks offline	Check physical power and connectivity before attempting any service actions.
Storage usage seems high	Check which storage is selected and what content types it supports; do not delete data.
VM vs CT confusion	Look at guest icon and inspect summary: VM uses its own guest OS kernel; LXC shares host Linux kernel.

7 | Vocabulary review

Bare metal: Physical server hardware, rather than an application running on a desktop OS.

Host: Physical machine running the virtualization platform.

Guest: Operating system or workload running in a VM or container.

KVM: Kernel-based Virtual Machine: Linux virtualization technology used by Proxmox.

LXC: Linux Containers: operating-system-level virtualization.

Resource tree: Left-hand interface listing available nodes, guests and storage.

Datacenter: Management root for resources and configuration.

Thin provisioning: Allocating storage on demand from a logical thin pool; requires monitoring.

ISO image: Installation media commonly uploaded to supported directory-like storage.

Cluster: Group of Proxmox nodes managed together using clustering capabilities.

Quorum: The voting majority needed for coordinated cluster operations.

Backup: Recoverable data copy of a guest or its configuration maintained according to policy.

8 | Review questions

■ What distinguishes a VM from an LXC container?

_____

■ Where can you find a node’s CPU and memory usage?

_____

■ Why do local and local-lvm generally store different content types?

_____

■ Why should technicians inspect rather than modify a new production environment?

_____

Official reference material

Proxmox VE Administration Guide (9.x): proxmox.com/en/downloads/proxmox-virtual-environment/documentation

Proxmox VE downloads: proxmox.com/en/downloads

Training diagrams and mock GUI are original instructional illustrations. Real installations may differ by version and storage configuration.